

2019 UX Global Meet

User Research Experience Sharing Workshop

version 1.0

Samsung SEA Product Experience

April 29, 2019

RESEARCH THINKING (POST – 2009)

- ❖ An Archaeology of Pre-Smart Multi-modal UX Design
- ❖ Flexible & Minimal Attention
- ❖ Multimodal User Interface Design
- ❖ Design of Future Things

- ❖ ISO Standard and Practice
- ❖ Metrics – Objective & Subjective
- ❖ QWERTY: Iterative Prototyping + Physical Design and Ergonomics
- ❖ GSIII Usability: Walkthrough + Performance-Based Usability
- ❖ Active vs. Passive Methodology
- ❖ Numbers can be deceiving

DESIGN THINKING (PRE-2009)

“Our knowledge can only be finite, while our ignorance must necessarily be infinite.”

~ *Karl Popper*

“Experts who acknowledge the full extent of their ignorance may expect to be replaced by more confident competitors, who are better able to gain the trust of clients. An unbiased appreciation of uncertainty is a cornerstone of rationality—but it is not what people and organizations want.”

~ *Daniel Kahneman, Thinking, Fast and Slow*

DESIGN & RESEARCH THINKING

**Provide framework for Expert Review and
Heuristic Evaluation of a product or service**

**Guide Usability and Product
Experience Research Protocol
Development**

DESIGN THINKING

**Multi-modal User Interface Design Guiding
Principles**

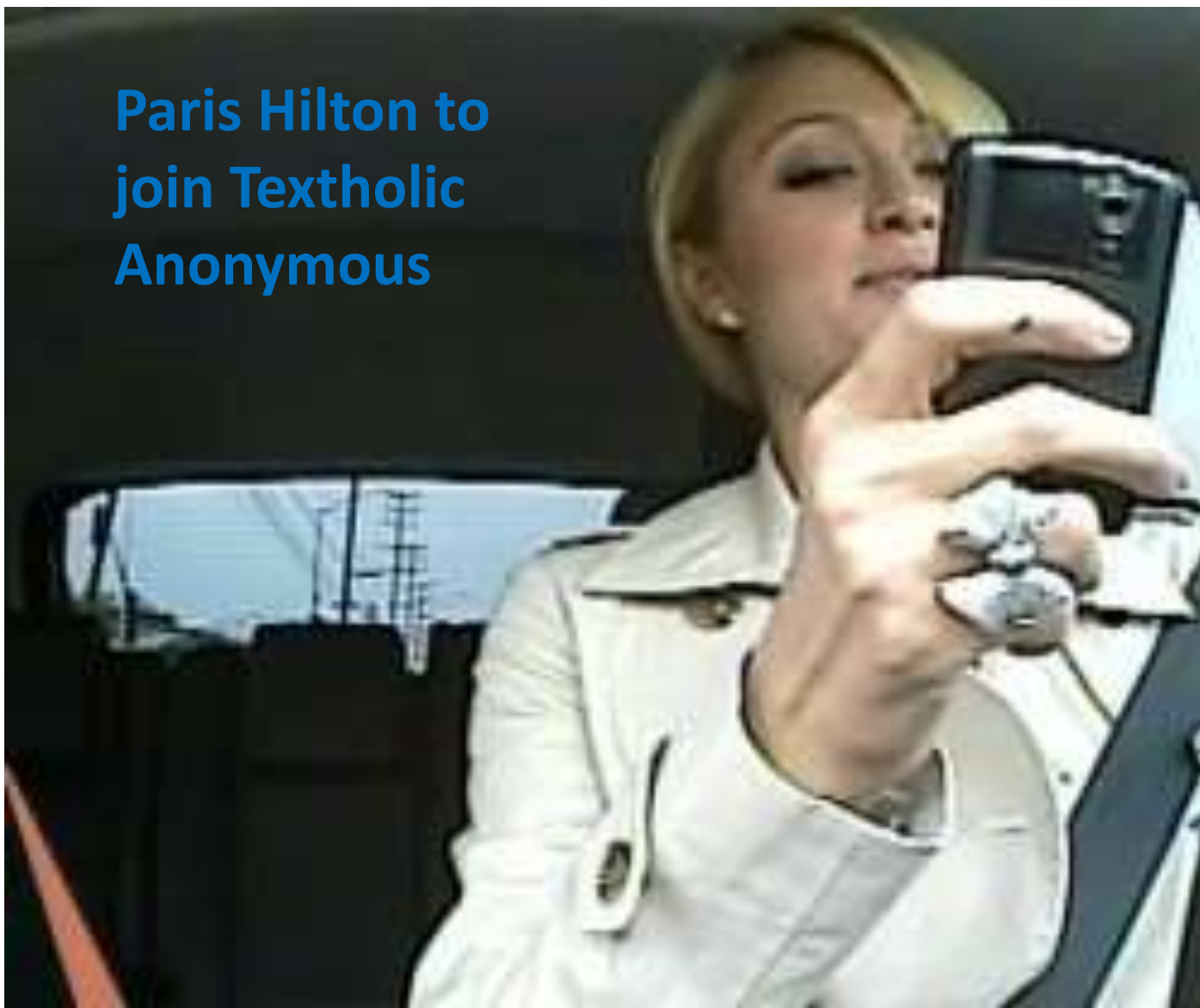
**A historical parallel to Smart
Display User Interaction
Design 2018/2019**

AN ARCHAEOLOGY OF PRE-SMART MULTI- MODAL UX – 2001 THRU 2008



England may be the first country to put pads on lamp-post to prevent texting-related injury

Paris Hilton to
join Textholic
Anonymous



Texting and Cycling don't mix



HOW OUR REACTIONS SLOW

This chart shows the percentage increase in distracted drivers' response times. An undistracted driver typically reacts in 1 second

13%



Drink-drive limit

21%



High on cannabis

27%



**Hands-free
phone**

37%

Texting



46%

Hand-held phone



Source: Transport Research Laboratory

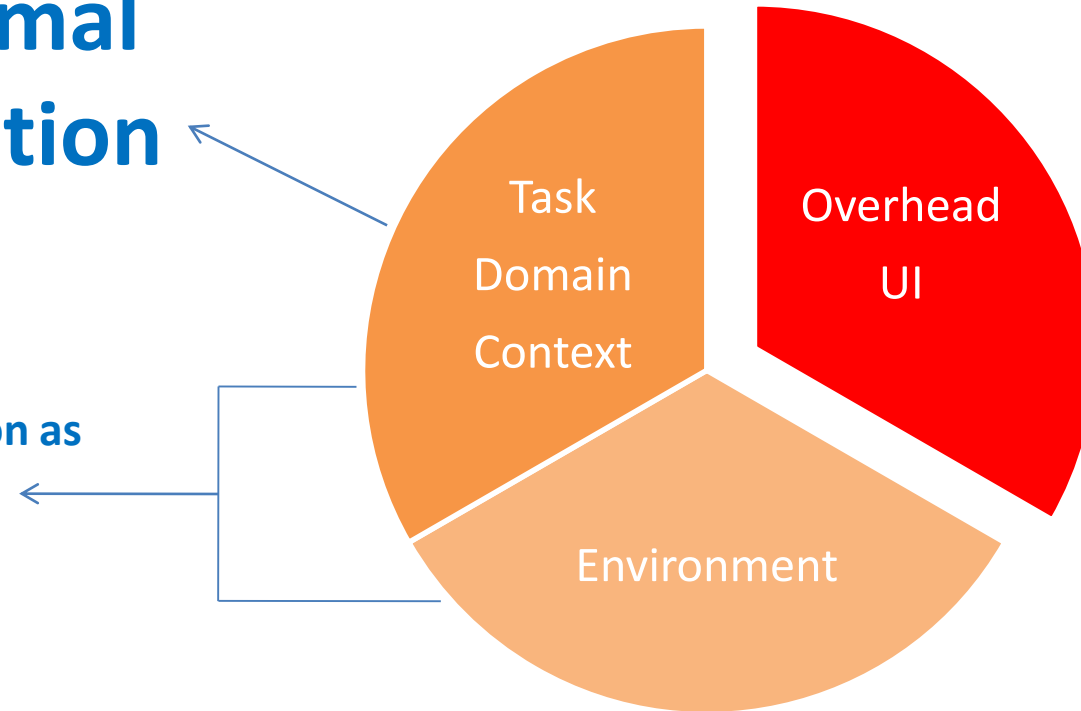
**focused, sustained and selective attention for
completion of tasks at hand, either work,
communication or entertainment**

**divided attention between
tasks and the changing
environment**

FLEXIBLE ATTENTION PRINCIPLE

**Maximal
attention**

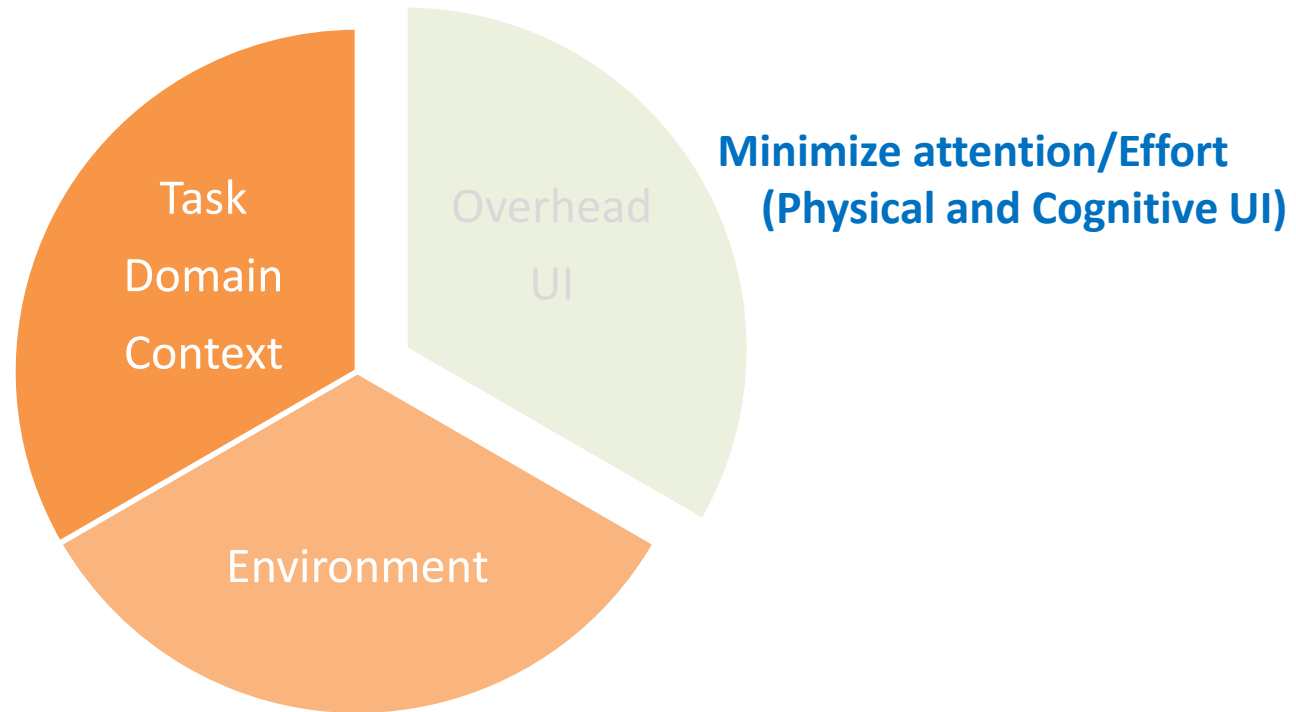
**Flexible attention as
necessary**



User Interface Design (cognitive and physical)
should minimize attention and effort to
operate the device (especially in mobile
environment).

Device user interface should disappear in the
background and user interactions should
become automatic without “thinking”.

MINIMAL ATTENTION PRINCIPLE



Steve Krug

**DON'T
MAKE
ME**

THINK



Flow is a finely tuned sense of rhythm, involvement, and anticipation. It is a state of "intense emotional involvement" and timelessness that comes from immersive and challenging activities such as software coding or rock climbing.

- Mihaly Csikszentmihalyi

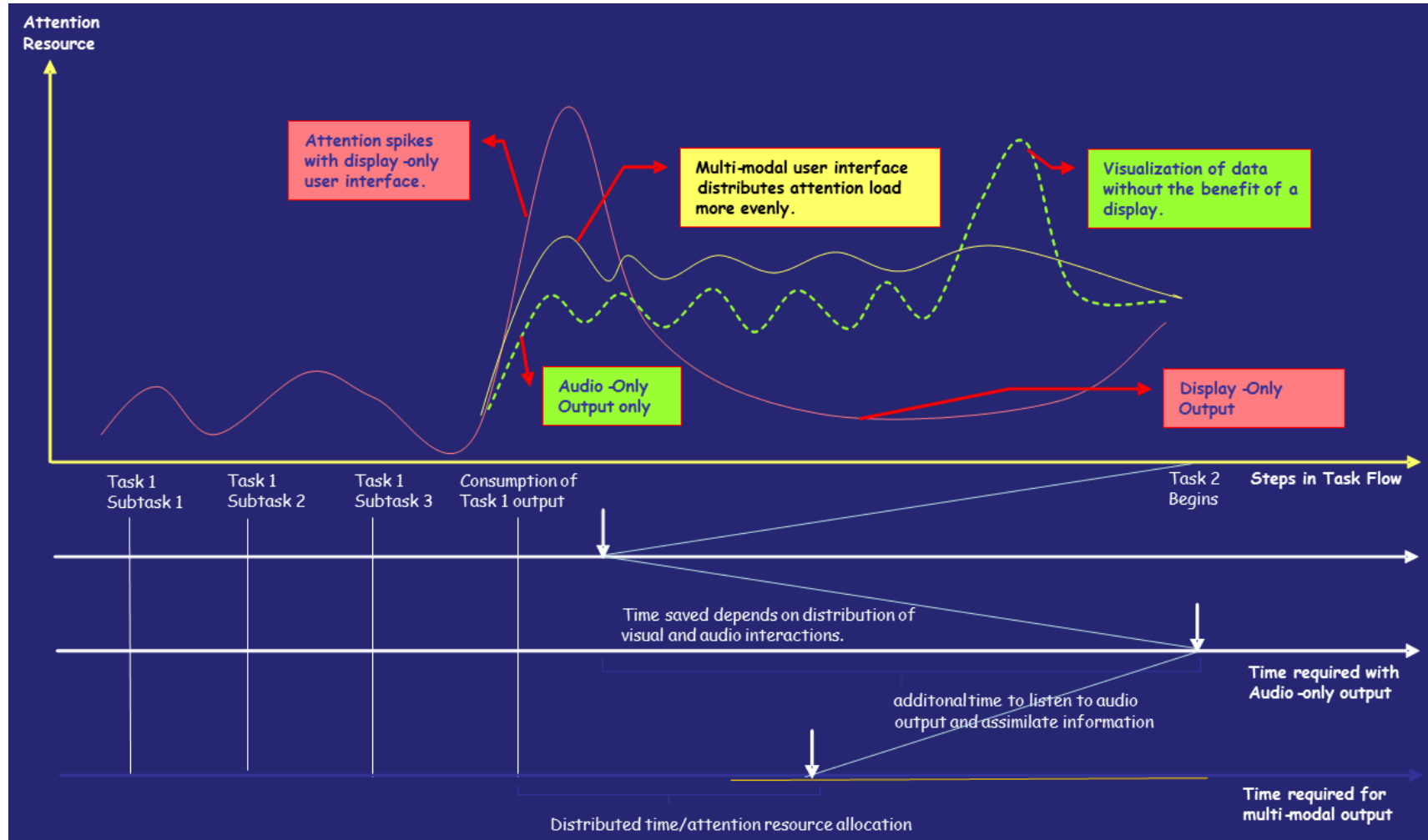
FLOW EXPERIENCE – KEY TO USER NIRVANA

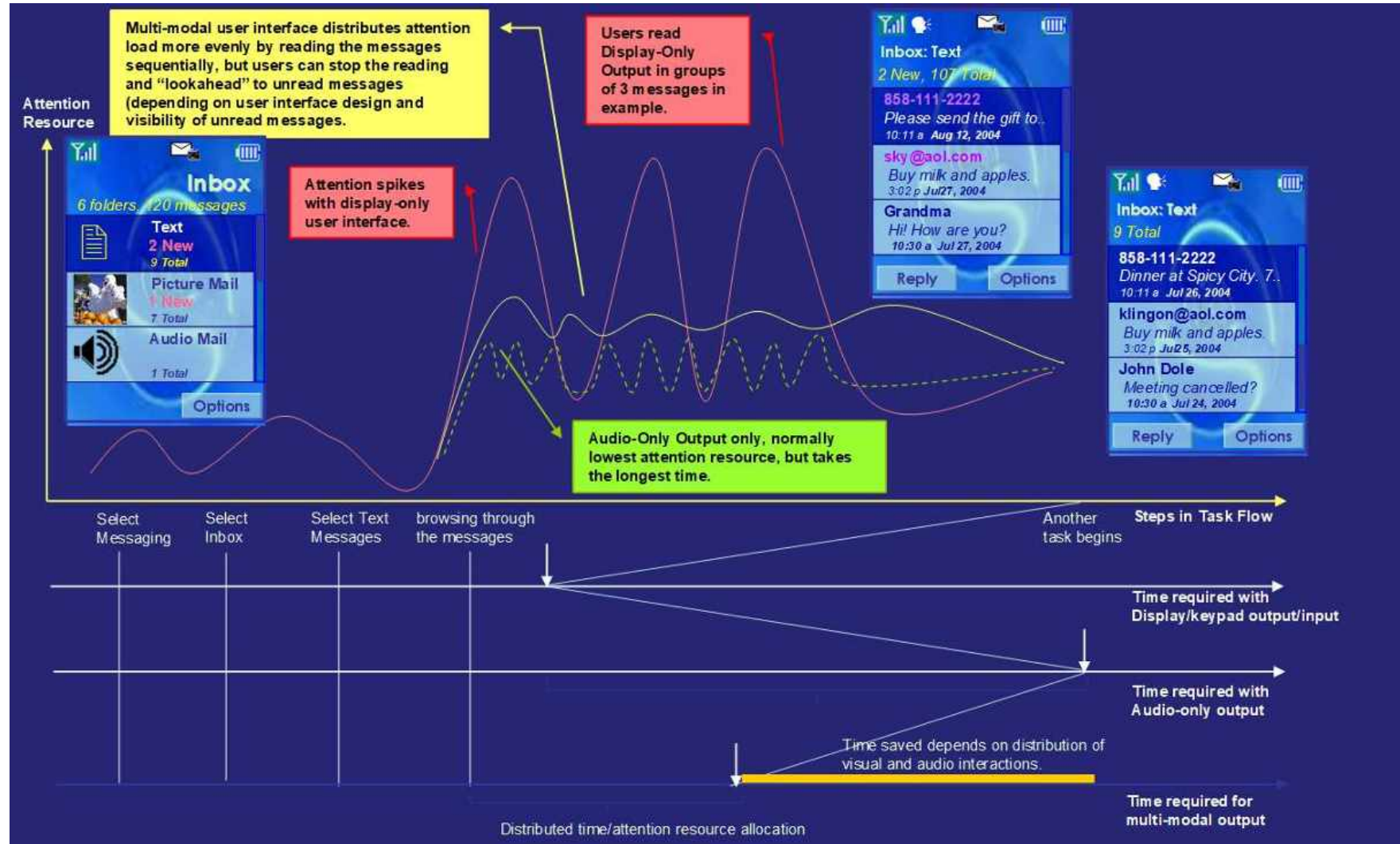
Multimodal User Interface Integration

**Dedicated Voice
Command Button**

SPEECH ENABLEMENT

**A COLLABORATIVE DESIGN EXERCISE – NO
POWERPOINT NEEDED**





The Design of Future Things (Donald Norman 2007)

- Design Rules for Human Designers of “Smart” Machines.
 - Provide rich, complex and natural signals.
 - Be predictable.
 - Provide good conceptual models.
 - Make the output understandable.
 - Provide continual awareness without annoyance.
 - Exploit natural mappings
- Design Rules Developed by Machines to improve Their Interactions with people.
 - Keep things simple.
 - Give people a conceptual model.
 - Give reasons.
 - Make people think they are in control.
 - Continually reassure.
 - Never label human behavior as “error”

Possibilities...

A device that can sense both our physical and virtual environments on our behalf to accomplish flexible distribution of attention resources.

WHAT'S THE NEXT BIG THING?

Interaction Design
Informational Architecture
Visual Design

Task Flow and Usability
Attention Allocation
Aesthetics, Look and Feel are somewhat secondary

WHAT IS NEEDED IN RESEARCH?

“Measure what can be measured, and make
measurable what cannot be measured”
~ *Galilei Galileo*

RESEARCH THINKING

**Provide framework for Usability and Design
Evaluations of a Product or Service**

**Guide Usability and Product
Experience Research Protocol
Development**

RESEARCH THINKING (POST-2008)

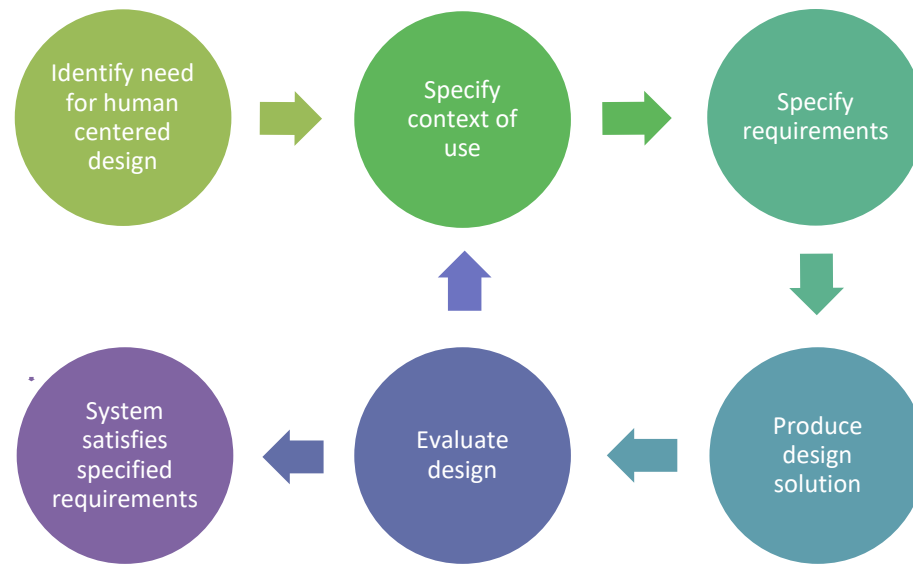
“In so far as a scientific statement speaks about reality, it must be falsifiable; and in so far as it is not falsifiable, it does not speak about reality.” ~ Karl Popper

GUIDING PHILOSOPHY

ISO 20282

Test methods for quantifying the usability of consumer products to ensure they meet a pre-defined quality level. (Recommended N =50 or above)





Using Human-centered Design Process (ISO 18529)

We approach product and user design with **User-Centered Design/Research Process**, whenever is possible

The process focuses on the needs and requirements of the actual users through a “Usability Maturity Model” defined by ISO 18529 throughout the planning, design, development and evaluation of the product.

Usability Slogans from Executive Summary

1. Your Best Guess is Not Good Enough
2. The User is Always Right
3. The User is Not Always Right
4. Users Are Not Designers
5. Designers Are Not Users
6. Vice Presidents Are Not Users
7. Less is More
8. Details Matter
9. Help Doesn't
10. Usability Engineering Is Process

~ Jakob Nielsen, Usability Engineering (1993)

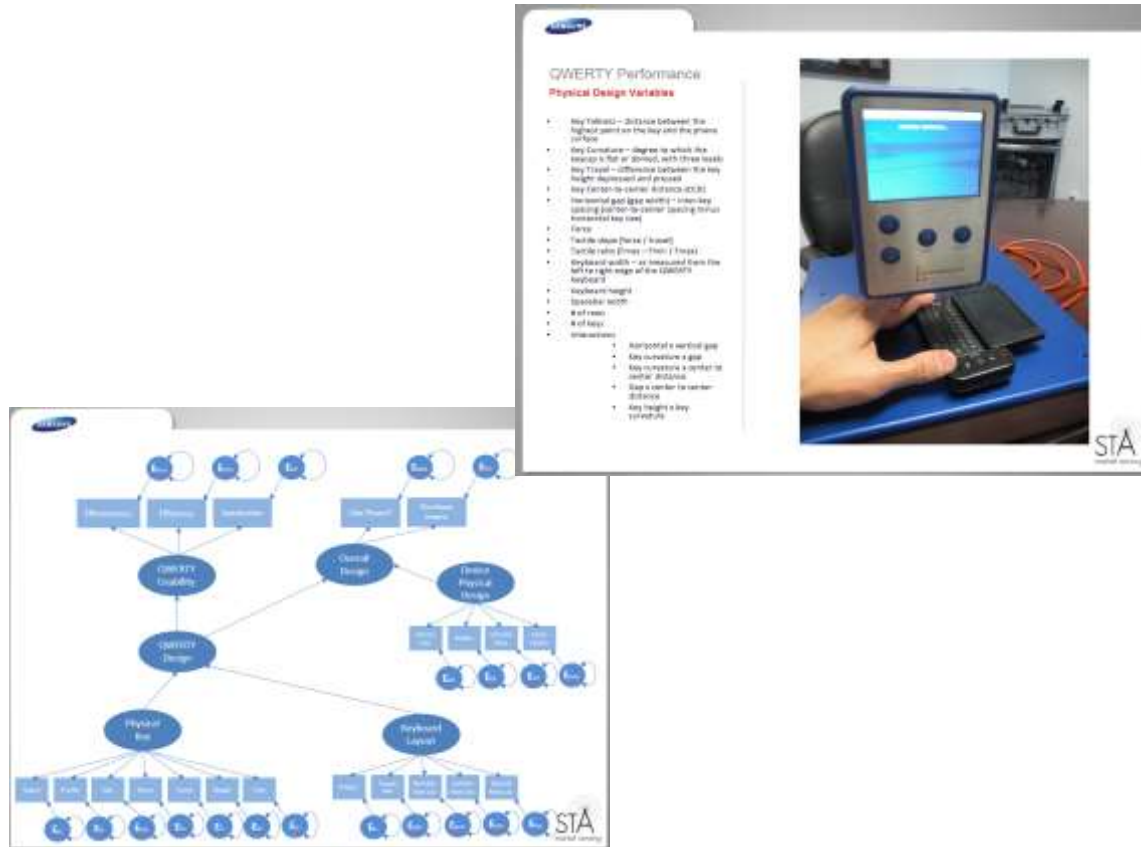
USABILITY ENGINEERING – JAKOB NIELSEN

SWAN SONG FROM USABILITY AS A SCIENCE

Large Scale Product Design and Ergonomics/Usability Research (Example)

- ❖ Best Practice Physical QWERTY Design for smartphones with QWERTY keyboard (2009)

COMPLEX LARGE-SCALE USER RESEARCH – CASE STUDY



HYPOTHESIS & USAGE MODELING

Use your intuition & domain knowledge to develop testable/measurable research hypothesis or user performance model. Define the appropriate metrics.

Objective Usability Metric

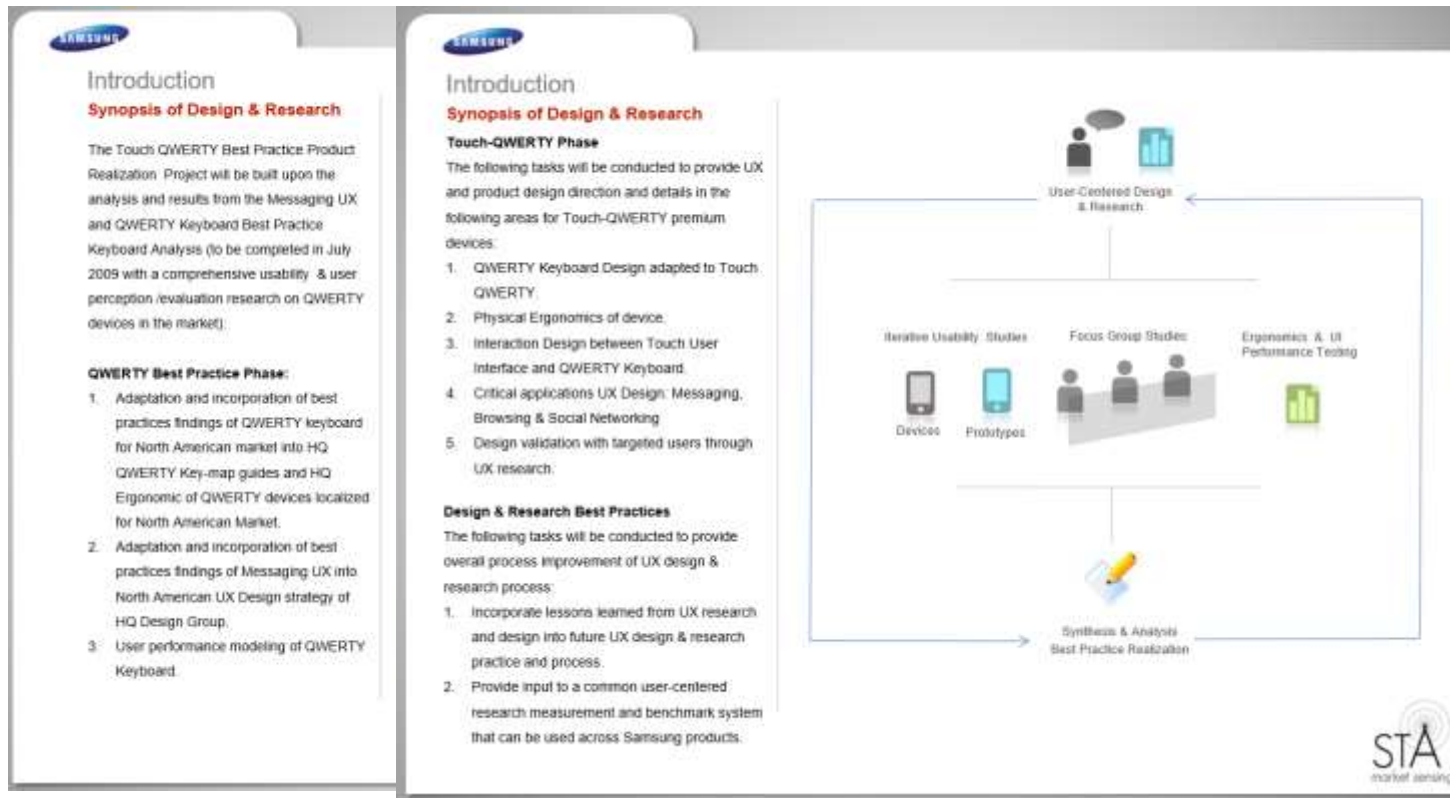
- Number of errors
- Speed (characters per minute)

Subjective Usability Scores

- Ease of Typing
- Satisfaction of Typing
- Just-Right Scale or Likert Scale for selected independent variables
- Ranking / Preference

Physical Design variables (key)

- Tallness
- Curvature
- Travel
- Center-to-center distance
- Horizontal gap
- Force
- Tactile slope
- Tactile ratio
- Keyboard width
- Keyboard height
- Spacebar width
- # of rows
- # of keys
- Interaction Effects



METHODOLOGY AND RESEARCH PROTOCOLS

It was a multi-phase project with complicated design, starting with paper prototyping through large scale real devices physical ergonomics and performance with physical keyboard to performance testing with virtual keyboard layouts. Note that in any of the research, users will need to use and type on keyboards irrespective of whether they are real products/prototypes or paper.

Introduction
Synopsis of Study

A competitive usability study of Samsung 2009 QWERTY keypads was conducted to evaluate the spatial key layouts of 4 different configurations - Portrait Keyboard, Landscape Keyboard, 3-Line, 4-Line and 5-Line QWERTY. Paper prototypes were used in the study to isolate the effects of key grids from the physical characteristics of the keys (e.g., trip force, tactile etc.). Devices, those are no Samsung devices with the key grids in the market for testing with real devices.

Paper prototypes of 4 Samsung QWERTY layouts were compared against the paper prototypes of the layouts of 4 competitive devices (PalmPre R500, LG E900, LG R900 and HTC Touch P320). Respondents also ranked the 4 benchmark competitive devices after the tasks.

STÅ
market sensing

Samsung QWERTY Guide
Results: Test Entry Performance

The following performance measurements were used in the evaluations:

- 1) time to complete task
- 2) Can the user learn the jump over the step?
- 3) ease of typing
- 4) ease of finding apostrophe (')
- 5) ease of finding question mark (?)
- 6) ease of finding @ symbol (@)
- 7) satisfaction of digital entry

A. Samsung QWERTY 2009 design keypad paper prototypes performed as good as or better than competitive benchmark keypad layouts within the four categories (Portrait, 4-Line Landscape, 3-Line Landscape, 3-Line Landscape) tested.

Samsung 2009 QWERTY Key-Grids:
provided a good and solid foundation to enhance user experience for future QWERTY keypad layout for North American market.

QWERTY Form Factor
Paper Prototype Preference

Paper Prototypes Results

Differences in ranking not statistically significant

The Paper Prototypes were constructed to isolate the influence of physical key characteristics (e.g., trip force, tactile response, material) from the layout of the QWERTY Keyboard (e.g., spatial locations and relationships of the keys, size of keypad and keys).

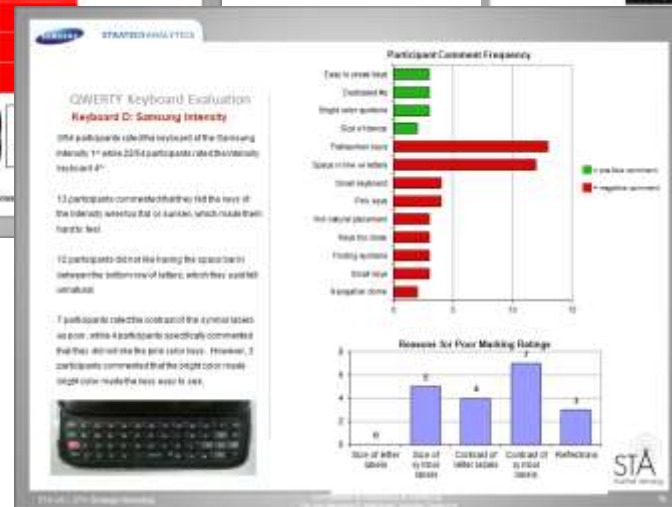
The ranks of LG R900, Samsung 3-Line, and Samsung 4-Line Paper Prototypes (7, 8 and 9) were significantly higher than the prototype of the QWERTY Portrait group. The visual key arrangement of QWERTY Portrait devices were contributing factors to the lower preference in this group as the size of keyboard was affecting effects on the legibility of the labels of the keys. In addition to the increased chance of engaging with a small keyboard.

The paper prototypes, LG R900, Samsung 3-Line, Samsung 4-Line ranked significantly higher than both PalmPre and Samsung Portrait. In addition, Samsung 5-Line and HTC P320 also ranked significantly higher than Samsung Portrait. If market's test at alpha = 0.05.

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PAPER PROTOTYPING AND REAL DEVICES BENCHMARK

Isolate the influence of physical key characteristics (e.g., trip force, tactile responses, material) from the QWERTY layout (e.g., spatial locations and relationships of the keys, size of keypads and keys etc.)



PHYSICAL DESIGN AND ERGONOMICS BENCHMARK

Use existing devices with different physical key design characteristics in the market to benchmark performance (both objective and subjective metrics) with different physical key design characteristics.

Research
Secondary Research

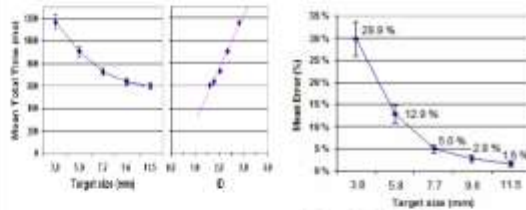
The following studies, together with the primary research and measurements of physical characteristics of the best performing devices (BlackBerry P and Tour), is used for the recommendations of button sizes and the micro-geometry of the asymmetrical button layout.

Karlson, A. K. (2007) Interface and Interaction Design for One-handed Mobile Computing. Ph.D dissertation.

Kalle, D. (2008) One-handed thumb use on smart phones by semi-literate and illiterate users in India. A usability report with design improvements for precision and ease. Workshop on Cultural Usability and Human Work Interaction Design. NordCHI Conference, Lund, Sweden, 2008.

Serina, E.R., Mole, C.D. & Rempel (1997). Force response of the fingertip pulp to repeated compression: Effects of loading rate, loading angle and antisymmetry. Journal of Biomechanics 30, 1035-1040.

Weinstein S. (1968). Intensive and extensive aspects of tactile sensitivity as a function of body part, sex, and laterality. In D.R. Kenshalo (ed.). The Skin Senses.

Target Size Study: The mean task time between the release of the start button and release of the target 'x' for each target size in the discrete target phase (left), and the relationship between movement time and task index of difficulty (right).

Target Size Study: The mean error rate for each target size

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Research & Analysis
Human Psychophysics

The two point touch threshold of fingertips is between 2mm and 4mm (Weinstein 1968). Thus, the minimum "center-to-center" distance between 2 buttons must be at least 4mm assuming the buttons have very distinct and pointed "peaks". Since the crest of each button is not a needle, the practical minimum distance depends on the changes of the gradient between the adjacent peaks defined by the buttons.




The recommended minimum length of the diagonal line of the key is 7mm with the optimal size to be around 10mm (a better curve to use is a "sweeping curve" across the between the diagonal and the lengths formed by the width and height of the key (see previous page).

The recommended length ranges of the "thumb guide" are extrapolated from touchscreen button size study (Karlson, 2007; Kalle, 2008).

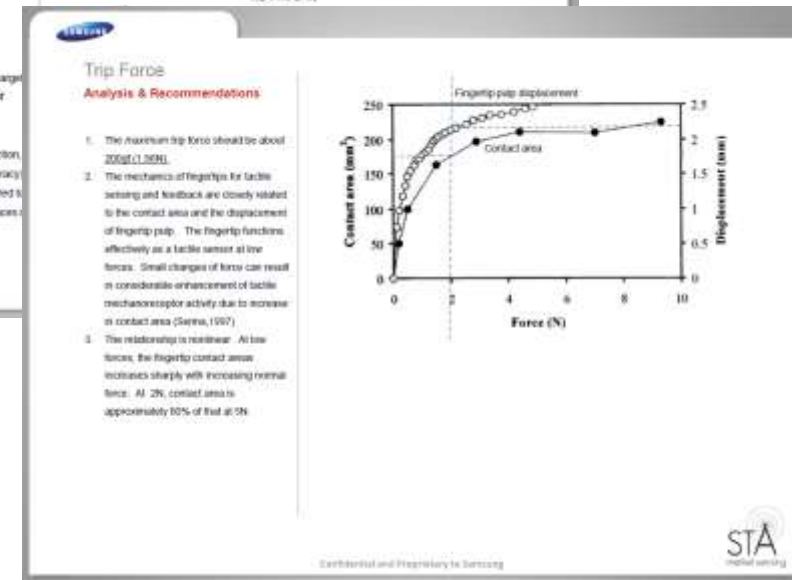
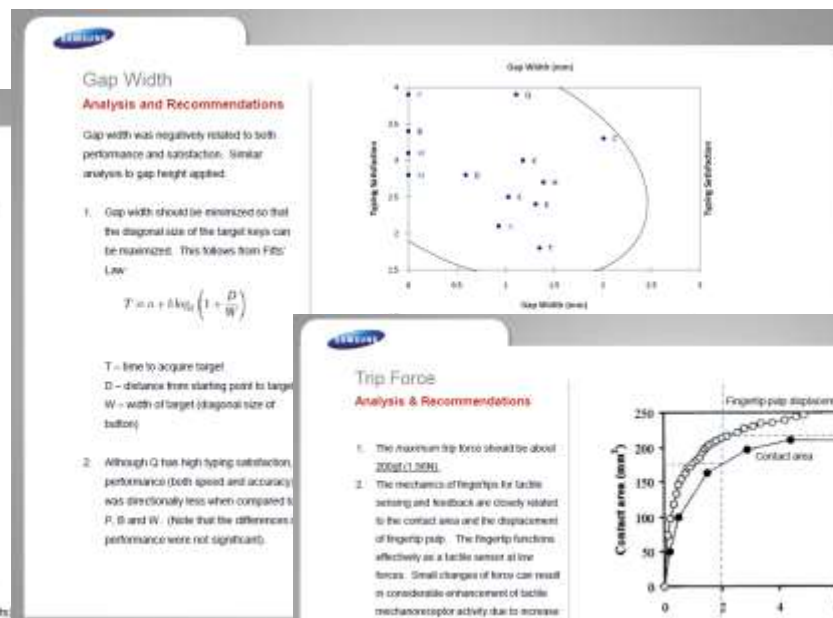
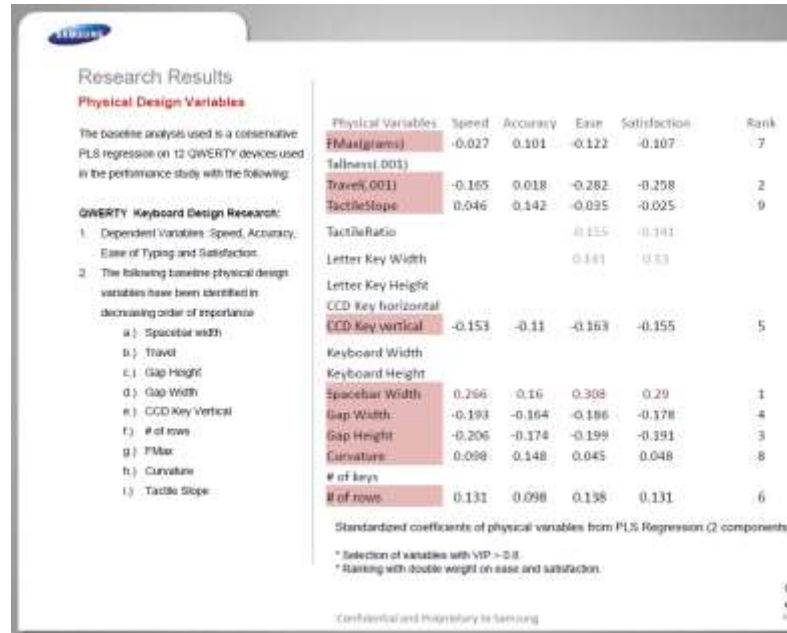
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IMPORTANCE OF SECONDARY RESEARCH & COMPETITIVE BENCHMARK

Secondary research can yield much deeper understanding and knowledge about and Avoid "Not Invented here" mentality.



ANALYTICAL METHODS AND INTERPRETATIONS

The appropriate analytical method(s) should be used. In this study, we used Partial Least Square Regression instead of Multiple Regression (which is simpler to apply and interpret). Use of PLS regression enables us to separate out co-variation

Summary			
Synopsis of Recommendations (Portrait)			
Variables	Lower bound	Recommended Range	Notes
Spacebar width	14mm	14.8mm – 21mm	Based on Blackberry Tour & LG W
Travel (0.001")	0.011"	0.014"	Best performing QWERTY devices
Gap Height	0 mm	0 mm – 1 mm	Curvilinear keyboard rows may need visual row separator to guide spatial location of keys
Gap Width	0 mm	0 mm – 1.2 mm	Lower is better to increase size of key
Center-to-Center Vertical Key Distance	4 mm	6 mm – 7 mm	4mm based human psychophysics
# of rows	4	4	4 rows standard design
Fmax (gf)	-	under 200gf	200gf from both primary & secondary research
Curvature	High	High	High for Portrait QWERTY
Tactile Response Slope	-	980-1100 gf/mm	Newly defined variable from ASTM standards
Number of Keys	35	35 - 37	35 keys recommended
Key Height	6mm	>6.5mm	Diagonal>7mm
Key Width	5mm	>5 mm	Diagonal>7mm

RECOMMENDATIONS – DESIGN

Recommendations were driven from the results of a series of empirical studies of typing performance with different QWERTY physical design parameters. They were used as a global physical QWERTY keyboard design guideline by HQ Design Center.

The overall research process should be rigorous and meaningful enough for a publishable paper (without proprietary specific recommendation disclosed).

Mobile QWERTY User Research

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137-857, Seoul, Korea

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Abstract. In this paper, we describe the methodology and the activities of mobile QWERTY research and give examples of different pieces of research and some of the results and insights. To minimize confounding effects of different aspects of QWERTY design, we use virtual keyboard layout tools to construct/replicate different keyboard layout designs so that both human performance and user satisfaction can be captured by systematically varying design parameters in the same testing environment. In addition, we also conduct usability testing by varying specific hardware design parameters and keeping all others constant whenever possible. By using this 2-pronged approach, an empirical performance model of mobile QWERTY keyboard in relation to different design attributes (both perceptual and mechanical) can be built over time. The approach we have adopted in virtual keyboard testing also helps to improve virtual QWERTY keyboard design on touch devices as well as physical QWERTY keyboard layout on devices with physical QWERTY.

Keywords: mobile QWERTY, human performance, user satisfaction, performance model.

1 Introduction

Usability of mobile QWERTY keyboard is determined by many different design variables: the overall size of the keyboard, the number of lines, the size, height, trip force, travel of individual keys, the geometry of key shape etc. Samsung Design Group at Korea and Samsung Telecommunications America jointly conducted systematic QWERTY ergonomics and usability evaluations of existing as well as prototype mobile devices to unravel the relationship between different design variables and establish best practices of mobile QWERTY keyboard design.

1.1 Research Goals

The primary goal in this program is to guide the development of new QWERTY devices by best practices informed by research and to test and improve new QWERTY keyboard design prototypes in development. These include the design and development of both physical and virtual QWERTY keyboards. By benchmarking typing performance, subjective user satisfaction and assessment of usability, we will continually improve our design recommendations and best practices.

A. Marcus (Ed.): Design, User Experience, and Usability, Pt I, HCI 2011, LNCS 6769, pp. 583–592, 2011.
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- ❖ Most, if not all the findings and recommendation/guidelines are still relevant to physical QWERTY devices in 2019.
- ❖ The metrics, research protocols and methodologies have been re-used in other important research, e.g., the MSD (Minimum String Distance) Excel tool to score typing errors has been used to compare dictation accuracy for voice assistants on smartphones as well as physical QWERTY research on laptops



BB KEY2 (\$699) launched April 19, 2019 by TCL

“...this is a no-brainer if you must have a phone with a physical keyboard” ~ Hardee Simons, Android Authority

<https://www.androidauthority.com/blackberry-key-2-red-edition-usa-977775/>

- ❖ Performance-based usability research is the golden standard of usability testing.
- ❖ One-on-one interview within a controlled usability laboratory is usually done professionally (with or without a moderator in the same room).
- ❖ Testing can be done effectively in group sessions with post test mini focus groups to optimize cost/time-saving and benefits
- ❖ Include benchmark comparison in the context of competitors and/or previous design when possible.

PERFORMANCE-BASED USABILITY

Metrics

- ❖ Objective: Task completion Rate, Number of Trials, Time Required, Error Rate etc.
- ❖ Subjective: Ease of Use, Efficiency and Satisfaction
- ❖ Navigation Path and Error Analysis

Example Tools

- ❖ Use actual device or app whenever possible
- ❖ Testing Interactive Design: InVision Prototype
- ❖ Testing Information Architecture: Optimal Workshop TreeJack

PERFORMANCE-BASED USABILITY

- ❖ Users perform usability tasks with real products or prototypes (electronic or paper)
- ❖ Moderator walkthrough demonstrates key features or functions of the product
- ❖ Participants to replicate what is demonstrated as tasks

TASK-BASED VS. WALKTHROUGH USABILITY RESEARCH

FIRST-HAND EXPERIENCE IS CRITICAL

Summary Results & Research Insights

Galaxy SIII Smartphone: Initial Setup and Usage of Core and Differentiable Product Experience

Overview

The purpose of this research was to identify UX improvement opportunities of the differentiable product experience of Galaxy SIII. It was conducted in late June 2012 at San Diego (JY22).

Tasks

1. Respondents interacted with a Galaxy SIII Self-Guided Sales Station in a simulated retail environment, watched short videos for retail and used TecTiles to launch and watch videos customized highlighting differentiable product features of sharing. They evaluated the usability of interactive demonstration station as well as completed product concept evaluation of Galaxy SIII.
2. Respondents attended a sales demonstration session highlighting differentiable product features. They evaluated the responsiveness and usefulness of the highlighted features.
3. Respondents were given Galaxy SIII smartphones and performed initial Gmail account setup and exploration of core features/applications. They evaluated the usability and ergonomics of the device.
4. Respondents performed a series of initial setup and usage tasks of differentiable product experience related to S-Beam, Information Ticker in locked screen, Voice command for camera, Share Photo, All Share Play, ScreenLock and TecTiles. They evaluated usability and importance of those features.
5. Respondents completed System Usability Scale and product concept evaluation of Galaxy SIII smartphone.

Results

Although the reactions to Galaxy SIII smartphone were highly positive, there were critical usability issues related to initial setup of some of the differentiable product features. The high failure rates of discovery or initial usage would likely cause substantial frustrations for new owners of Galaxy SIII. Besides, those differentiable features might never get used and the owners would not get the full benefits of Galaxy SIII.



Most problems in the initial setup were related to the discovery of settings or usage of those product features. Many of them were hidden in unanticipated options in settings. Most users, for instance, would look for Screen Lock setup for locked screens under "Display". Many of them would also look for apps or widgets related to news. They did not anticipate that the setup option would be under "Security". For the voice command to take pictures, most respondents would first search for the settings within the camera application. They would also search "Sound" and "Accessibility" in the "Settings" menu and not expect the option would be found in "Language and Input".

Recommendations

Opportunities for improvements of Galaxy SIII included the following:

Short-term

1. Turn on some of the differentiable product features by default, e.g., S-Beam can be set on by default.
2. Organize the settings menu in a more logical/natural way according to the highlighted product experience, e.g., use a "Sharing" subtab under Settings to organize.

Long-term

1. The option to turn specific features on/off related to an application should be accessible within the application itself, e.g., turning on and off voice command for taking pictures should be accessible from the camera.
2. A widget or app can be used to help users to customize their smartphones, including direct access to the highlighted differentiable features and help/resources to set up and use those features.



Galaxy SIII Interactive Sales Demo Station (NFC/TecTiles) User Experience

Self-Guided Tour of Galaxy SIII

Results:

More than 90% of consumers enjoy the interactive experience of the station and think that the videos are informative. The experience can be further improved with explicit instructions of using TecTiles to launch videos.

Respondents were asked to walk up to the self-guided interactive sales station and explore the exhibit for about 5 minutes. Respondents would be prompted to make use of the TecTiles to launch videos of the corresponding "sharing" usage highlighted in the exhibit if the respondents did not do that after 2 or 3 minutes of exploration.

Although most respondents thought that they could walk through the interactive station without assistance, it took respondents more than 1 trial to get the Galaxy SIII smartphone attached to the security device working with TecTiles to launch videos for specific features.

Comments from the respondents suggested that the diagrams shown on the exhibit did not show how to place the device on TecTiles accurately. They were also confused by the name "TecTiles" and did not know what they were for. More explicit instructions and accurate depiction of how to place handset on TecTiles will be helpful.

Overall, the experience of consumers to the interactive sales station is very positive.



The picture should show placement of a Galaxy SIII attached to security cradle on TecTiles. The location of the back of device for placement should also be marked on the actual demonstration unit.

Diagram should show back device touching back to back to ensure NFC connection. It needs demonstration side clarity and difficult because of the security attachment.

More explicit instructions should be given to place smartphone on TecTiles. A brief description and explanation of TecTiles should also be given.

Galaxy SIII Sales Demonstration and perceived differentiable product experience

Consumers' Perception of galaxy SIII after Salesperson Demonstration

Results:

Consumers are impressed with the sales demonstration of features and agree that it takes minimal effort to understand important features of Galaxy SIII and it appears to be easy to use. They also agree that Galaxy SIII stands out among the best smartphones they have had seen.

The following features were demonstrated: S-Beam, Burst/Best Shots, Direct Call, Smart Stay, Share Play (streaming to PC using a HD TV as monitor), TecTiles (connecting to Pandora), and People Play. After the demonstration, 14 out of 23 respondents indicated that they would purchase Galaxy SIII while 8 would wait and compare Galaxy SIII with iPhone 5 and 1 for Blackberry RIM smartphone. Note the purchase intent results were not statistically significant.

Comments captured from consumers after the product demonstration showed that consumers, irrespective of the brand of their smartphones, were very impressed with the product features. Apple iPhone owners and insiders also indicated highly positive views of Galaxy SIII.

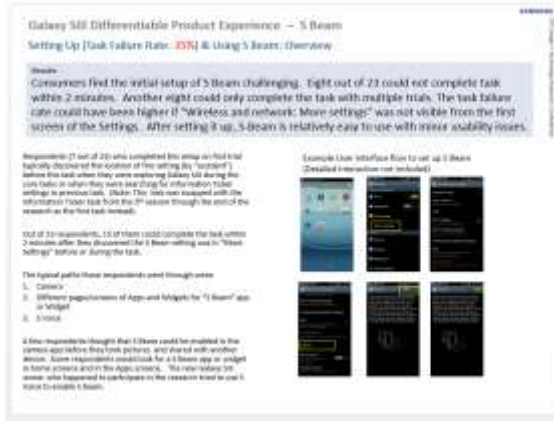


Feature	Order	Comments
Screen Lock	1	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	2	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	3	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	4	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	5	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	6	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	7	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	8	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	9	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	10	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	11	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	12	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	13	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	14	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	15	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	16	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	17	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	18	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	19	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	20	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	21	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	22	Most people are using the screen lock and they are not using the lock for the first time.
Screen Lock	23	Most people are using the screen lock and they are not using the lock for the first time.

GALAXY SIII USABILITY AND PRODUCT EXPERIENCE

Consumers used and evaluated the retail sales self-guided tour station (prototype) in simulated retail environment. They attended a “sales rep” live demonstration (scripted with official sales guide) of the differentiable features (alias USPs) and evaluated their first impression of the device.

They were then given a set of tasks to perform including physical design and ergonomics evaluation, core product experience (internet, messaging and dialer) and 9 different tasks related to the differentiable features demonstrated by the sales rep. Completion rates and usability scores were collected for each task.

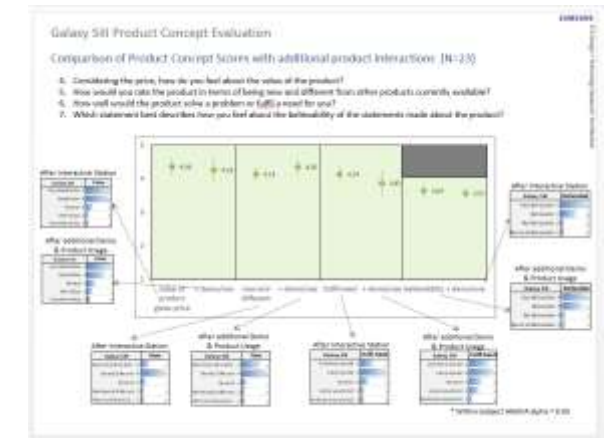
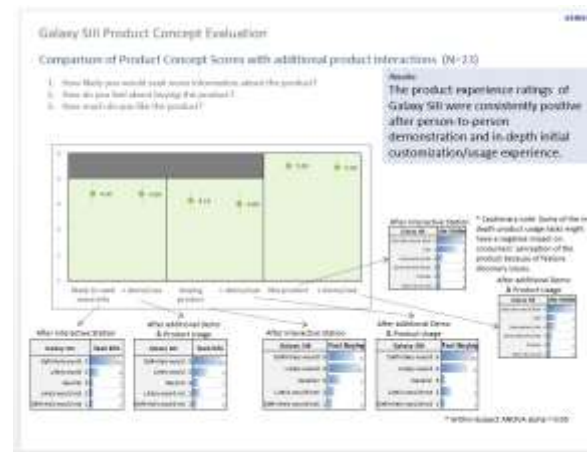
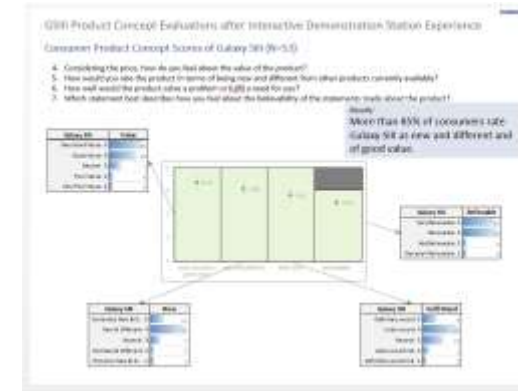
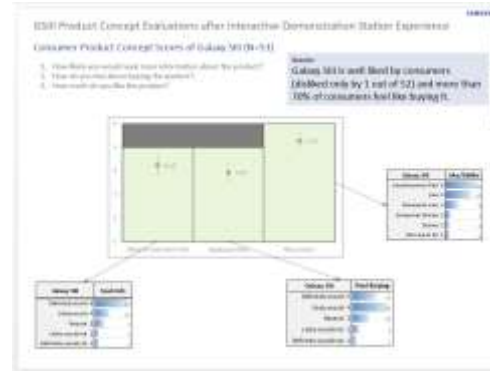
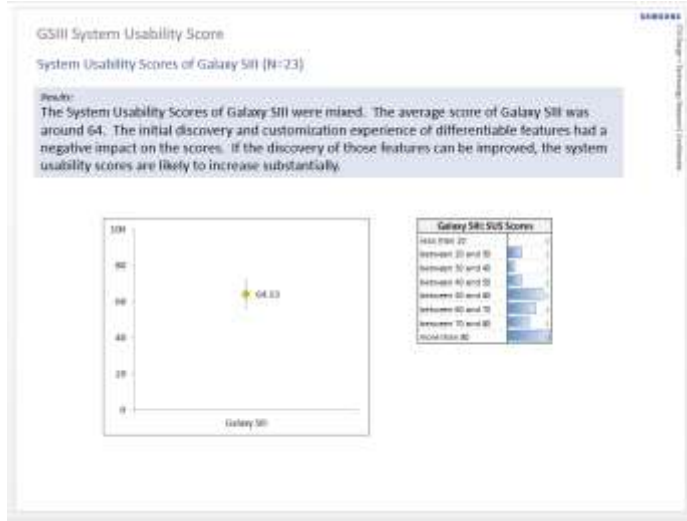


GALAXY SIII USABILITY – EXAMPLE TASK ANALYSIS

The tasks participants performed were:

- ❖ News Ticker
- ❖ S Beam
- ❖ Voice Command to take Pictures
- ❖ Editing Shortcut to Camera Menu
- ❖ Share Shot (Subtasks: Finding Share Shot, Turning on Wi-Fi Direct, Changing Device name or User Profile, Connecting with another device, Accepting Connections, Taking Pictures with Share Shot, Finding Shared Pictures),

- ❖ AllShare Play Photo Streaming to PC AllShare (Subtasks: Finding Media to stream, Finding AllShare Stream, Joining AllShare Play Groupcast, Completing Joining GroupCast, Changing Pen Color),
- ❖ Creating AllShare Play Groupcast (Subtasks: Finding AllShare GroupCast, Finding Location supporting Multiple Media, Multiple Selection of Media and Creating Groupcast)



GALAXY SIII OVERALL USABILITY AND PRODUCT CONCEPT

Product Concept Scores of Galaxy SIII Before and after hand-on interactions were also collected. These informed us the effectiveness of the self-guided demo station, the live demonstration of sales rep and actual product hands-on experience.

Galaxy SIII Differentiable Product Experience – S Beam

Setting Up S Beam: Discovery (Task Failure Rate: **35%**)

Results:
Consumers found the initial setup of S Beam challenging. Eight out of 23 could not complete task within 2 minutes. Another eight could only complete the task with multiple trials. Consumers would not anticipate S-Beam would need to be turned on before use.

Respondents did not anticipate S-Beam was turned off by default and had expected it would work out of the box.

Suggestions based on respondents' feedback:

1. A setup app or widget to customize the key differentiable features of the product.
2. Out-of-Box support for S Beam (default on) instead of "off".



Galaxy SIII Differentiable Product Experience – S Beam

Using S Beam: Usability

Results:
Consumers found using S Beam easy and straightforward except for minor difficulties in the onscreen instruction cue "Touch to beam". Respondents tried to touch the label instead of the object (e.g., photo) to initiate beaming.

One minor usability problem was the textual label of "Touch to beam" – many respondents would tap on the textual label itself multiple times, only finding that it did not respond and S-Beam timed out.

Suggestions based on respondents' feedback:

1. Provide explicit visual cue to the object from the label "Touch to beam" (e.g., "Touch picture to beam")



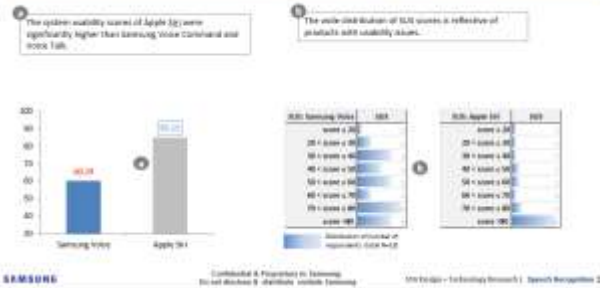
GALAXY SIII RECOMMENDATIONS FOR IMPROVEMENT

Product Concept Scores of Galaxy SIII Before and after hand-on interactions were also collected. These informed us the effectiveness of the self-guided demo station, the live demonstration of sales rep and actual product hands-on experience.

Automatic Speech Recognition – System Usability Scores

Samsung Voice talk / Voice command & Apple Siri

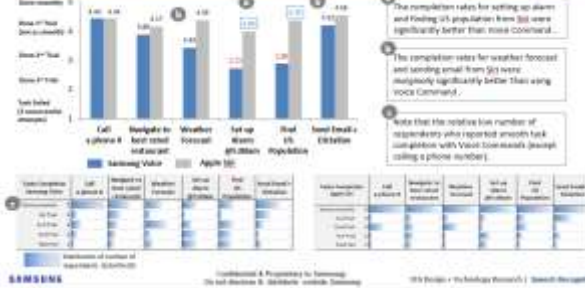
Insight:
The user experience of Samsung Voice Command/Talk substantially lags behind Apple Siri due to a variety of reasons. Among the most important factors are the rigid and limited vocabulary, lack of a single-step invocation of contexts and the lack of contextual understanding of speech input.



Task Completion - Overview

Task Completion Rate

Insight:
Consumers can complete more voice-enabled tasks with Siri than Voice Command and Voice talk. This can be attributed to different factors – the flexibility of Siri's semi-natural language input versus the rigid and limited commands of Samsung Voice as well as functional bugs in Samsung Voice.



Research Insight from Consumer Behavioral Research

Opportunities: Design Recommendations

Insight:
Direct and effortless invocation of front-end speech recognizer is the first (and easiest) product design direction – a dedicated physical button (and/or a simple gross motion-based gesture) should be included in competitive product if we decide to compete in this area.

Physical Design/Ergonomics

- The front-end speech recognizer can be invoked directly with a single button from both locked screen. A physical button is much more preferred to touch screen invocation. This can be a dedicated side button (physical).
- A motion-based gesture can be used conveniently to invoke speech recognizer instead of touching the display. The double-tap on Samsung S3 to wake up the speech recognizer is a viable approach.

User Experience Design & Speech Recognizer Natural/Artificial Intelligence Capability

- The development of natural/artificial intelligence knowledge base and capabilities are critical to compete effectively in contextual understanding. This is also the most difficult and challenging areas. Understanding of user behavior and designing products effectively to satisfy user needs require interaction of social/human sciences and engineering.

Application, Conversation & Dialog Design

- The experience design of Siri, including context and multi-modal interactions permeates with modern linguistic theories and conversation principles. Pursuing superficial design without knowledge in psycholinguistics and sociolinguistics will only lead to mediocre product experience.
- The Voice Action input should be clearly labeled in the widget and the widget can be renamed as Voice Action/Search. The current Voice Action features on Voice Search SLD is not discoverable.

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100 Design / Technology Research / Speech Recognition 10

Apple iPhone 4S Siri – a cursory Anatomy

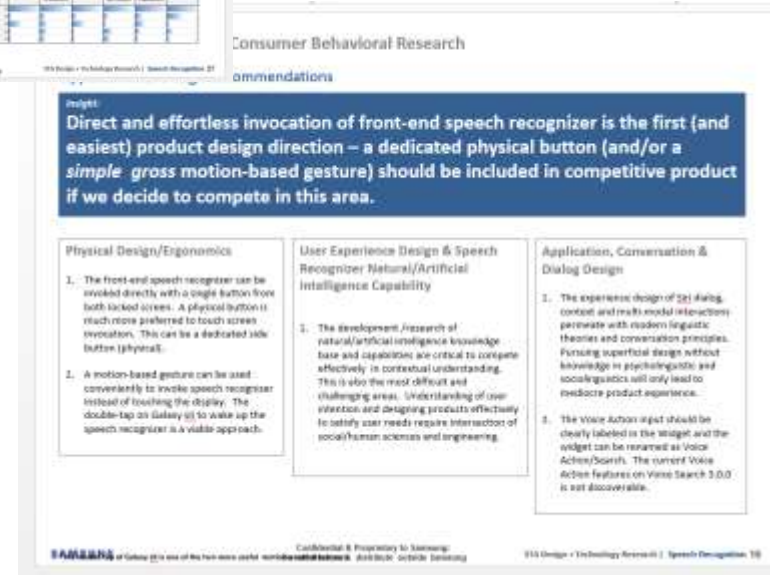
Invocation and Speech Recognition Applications

Insight:
Apple iPhone 4S Siri has a single but single step to invoke speech recognizer from virtually anywhere (any app) in the user interface. This design is the pinnacle of minimal but effective design. It minimizes both the attention and effort to invoke speech and makes its user interface transparent and trivial to learn. This also signifies the primacy of natural multimodal interactions in the vision of Apple for the future.



COMPETITIVE BENCHMARK 2011 – VOICE ASSISTANTS

Competitive Benchmarks for both short term improvements as well as long term strategic recommendations.



COMPETITIVE BENCHMARK – VOICE ASSISTANTS 2011

Competitive Benchmarks for both short term improvements as well as long term strategic recommendations.

23. Tablet Placements and Use: Understanding user needs and expectations

Overview

Space	Top Apps	Locations
Living Room	Facebook, Games, Music, Browser, Weather, TV shows, TV remote App	docked on table, flat on table, on couch, docked near TV, flat on a counter near TV, on a wireless charging pad
Kitchen	Music, Recipe websites, Cooking help sites, YouTube	docked on counter, kept on a stand, flat on couch, wall mounted, mounted on fridge, in a waterproof case on counter
Bedroom	Alarm clock, Music, Books, Movies, Note taking, Calendar	docked on nightstand, flat on nightstand, on nightstand as alarm, docked on wall next to bed, lying on bed, on dresser (near TV)
Bathroom	Music, Games	hanging on wall, on counter, next to pot, next to tub, in hand
Car	GPS, Music, Traffic updates, Maps, Movies, Hands free talk	on dashboard, lying on passenger seat, mounted on back of seat, on leg, mounted on steering wheel



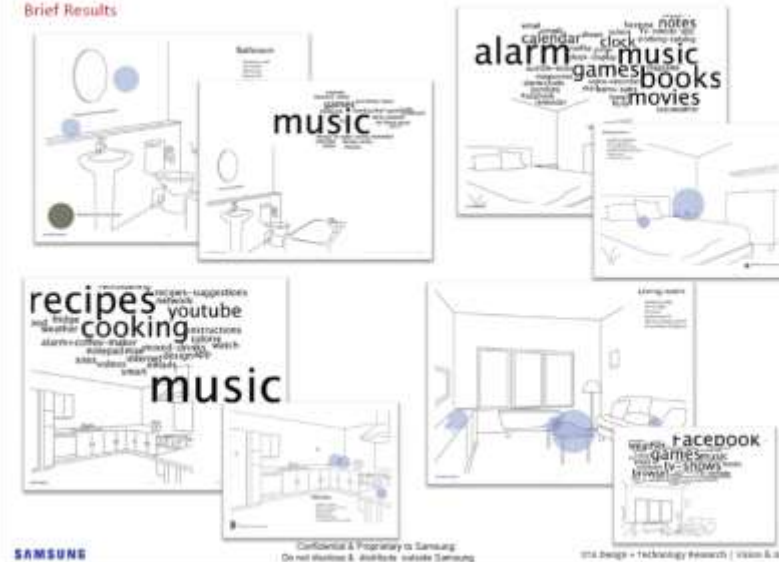
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STG Design + Technology Research | vision & scale 45

23. Tablet Placements and Use: Understanding user needs and expectations

Brief Results



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TABLET + DOCK PLACEMENT AT HOME 2011

Missed Opportunities: A precursor to Smart Display Placement and Apps

Limit Use of “traditional” focus groups in usability research

“Traditional” focus groups have been widely used in market research but is not an appropriate methodology to capture behavioral usability data. It typically has high entertainment (like a “live” talk show) but “low” usability value.

We can use “controlled” focus group with individual hands-on interactions (without discussion first)

Potential problems:

- ❖ Traditional focus groups are highly susceptible to group dynamics and dominant participants can influence the “opinions” of the group.
- ❖ Traditional focus groups are typically not “hands-on” research

PASSIVE VS. ACTIVE RESEARCH

“A simple rule can help: before an issue is discussed, all members of the committee should be asked to write a very brief summary of their position. This procedure makes good use of the value of the diversity of knowledge and opinion in the group. The standard practice of open discussion gives too much weight to the opinions of those who speak early and assertively, causing others to line up behind them.”

~Daniel Kahneman, *Thinking, Fast and Slow*

Avoid these pitfalls whenever is possible

- ❖ Prototype or device usability “comparison” with no hands-on participation. At the very least, participants should be following through (hands-on) with the product/prototype to be tested.
- ❖ Walkthrough Only Demonstration of Products and services with no subsequent hands-on evaluation
- ❖ Asking participants to give ratings or talk about usability issues of prototypes in a traditional focus group settings without completing their evaluation individually first

PERFORMANCE-BASED USABILITY

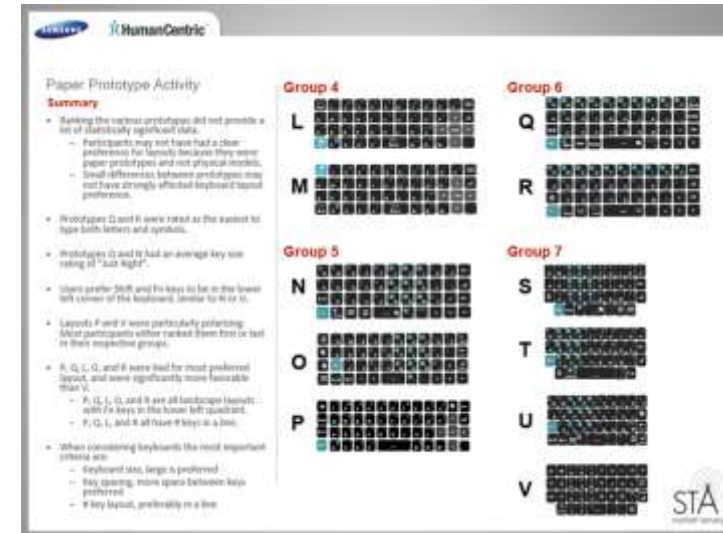
LESSONS LEARNED

- ❖ Paper prototyping exercises can work magic in early design
- ❖ Active participation (not the type in traditional focus group studies) is essential for all phases of product design and usability research
- ❖ Combine different methodologies in planning and execution of single research sessions to keep participants engaged
- ❖ Use the largest number of participants you can afford and analyze your pilot carefully
- ❖ Use realistic simulation (it is like having participants to be actors in a theatrical play) to capture needs and usage
- ❖ Make use of every opportunity to learn and run research related to topics of interest in parallel to primary research undertaken by vendor (e.g., using floaters in primary research or recruiting more participants to run parallel sessions)



2-in-1 Tablet/Laptop User Research 2017

- ❖ This is highly effective in different types of usability research including early product concept study, physical design and ergonomics, user needs exploration



QWERTY Keyboard Layout paper prototypes 2011

**ACTIVE PARTICIPATION: RESEARCH AS THEATER -
ACTIVITY-BASED PRODUCT DESIGN AND USABILITY RESEARCH
INCLUDING MOCKUP/PAPER PROTOTYPE**

- ❖ Constant stimulation leading to better engagement
- ❖ Re-enactment of usage for more valid, reliable and accurate product experience and usability assessment
- ❖ Physical Design & Ergonomics Evaluations, Usability evaluations,

RESEARCH SESSIONS WITH DIVERSE METHODOLOGIES

.

-

- ❖ Research results with number of participants less than 30
- ❖ Research results reported with only averages of objective and/or subjective usability scores without statistical testing (or at the least confidence intervals or error bars)

**RESEARCH RESULTS WITH TOO FEW PARTICIPANTS OR
WITHOUT ADEQUATE STATISTICAL ANALYSIS SHOULD BE
USED WITH SKEPTICISM**

STRATEGYANALYTICS

Executive Summary



- In a 'blind' user performance benchmark, the display quality of Samsung Super AMOLED + was rated significantly better than the HTC Evo, Apple iPhone 4 and Motorola Droid X
 - After watching a video clip simultaneously on all four devices, 192 out of 328 participants rated the Samsung Super AMOLED+ as their most preferred device for overall display quality.
 - Apple iPhone 4 was rated significantly higher than Motorola Droid X, which was rated significantly higher than HTC EVO
 - In a control sample, Apple iPhone 4 was rated significantly higher than Motorola Droid X, which was rated significantly higher than HTC EVO
- Samsung Super AMOLED+ rated significantly higher than all devices for all attributes participants evaluated except for color
 - Attributes evaluated were brightness, color, sharpness, smooth rapid motion scenes, view from different angles, picture quality, contrast and viewing experience
 - Samsung Super AMOLED+ rated the same as Apple iPhone 4 for color, but higher than HTC Evo and Motorola Droid X

Display Quality Benchmark: Samsung Super AMOLED + Display vs. Competitors



23 May 2011
Confidential

STRATEGYANALYTICS

A HIGH-STAKE MARKETING CLAIM

It was a marketing claim for the superiority of Super AMOLED on Samsung Galaxy compared to iPhone 4 and other Android devices. Legal approval with 2 independent companies advising methodology and carrying out the research were required.



STRATEGYANALYTICS

Introduction

- The goal of this study was to benchmark the display performance of the Apple iPhone 4 (Device A) against the Samsung Stealth Prototype (Device B)
- Strategy Analytics conducted a 'blind' user performance benchmark in San Diego, CA.
 - Both devices were hidden so the user could only see the display of each device – users were not told the make/model of either device.
- Total n=68 users
- Same video watched on each device



THE HIGH-STAKE CLAIM PILOT FROM HELL

When we ran the first pilot study with 68 participants, the first 20+ responses did not show any trend, which we knew was problematic.

The problem was traced to a subtle setup issue which confounded the study. Although we had hidden the front display of each device carefully. One side of the devices under comparison were somehow visible, leading to Apple iPhone owners unconsciously rating iPhone display higher (the infamous “brand” effect). When we hid the entire setup from the respondents, exposing only the front screen, the effect disappeared

- ❖ "A mythical bird that never dies, the phoenix flies far ahead to the front, always scanning the landscape and distant space. It represents our capacity for vision, for collecting sensory information about our environment and the events unfolding within it. The phoenix, with its great beauty, creates intense excitement and deathless inspiration."

~ The Feng Shui Handbook, feng shui Master Lam Kam Chuen

THE PHOENIX OF USABILITY SCIENCE AND ENGINEERING?

- ❖ Willing to experiment, take risk and fail
- ❖ Pay your due with real hands-on experience; Learn how to do E-2-E user research – Plan and run your own research, analyze and interpret data yourself
- ❖ Learn from professional researchers from different agencies; improve upon their methods
- ❖ Driven by curiosity and urge to learn and understand
- ❖ Research rigor is harder to find than “research” insights

RESEARCH THINKING - MINDSET

Tabla Rasa

ADDITIONAL MULTIMODAL DESIGN SLIDES (PRE-2008)

Automatic Speech Recognition

Speech recognition is the process of converting an acoustic signal, captured by a microphone or a telephone, to a set of words. The recognized words can be the final results, for such applications as commands & control, data entry, and document preparation. They can also serve as the input to further linguistic processing in order to achieve text formatting or speech understanding.

- Ron Cole 1997

Speech recognition is not perfect, human and machine alike.

Accuracy can vary substantially, depending on the type of task,
audio quality (both network, device, accessory),
speaker accents, ambient noise,
unpredictable changes in environment..



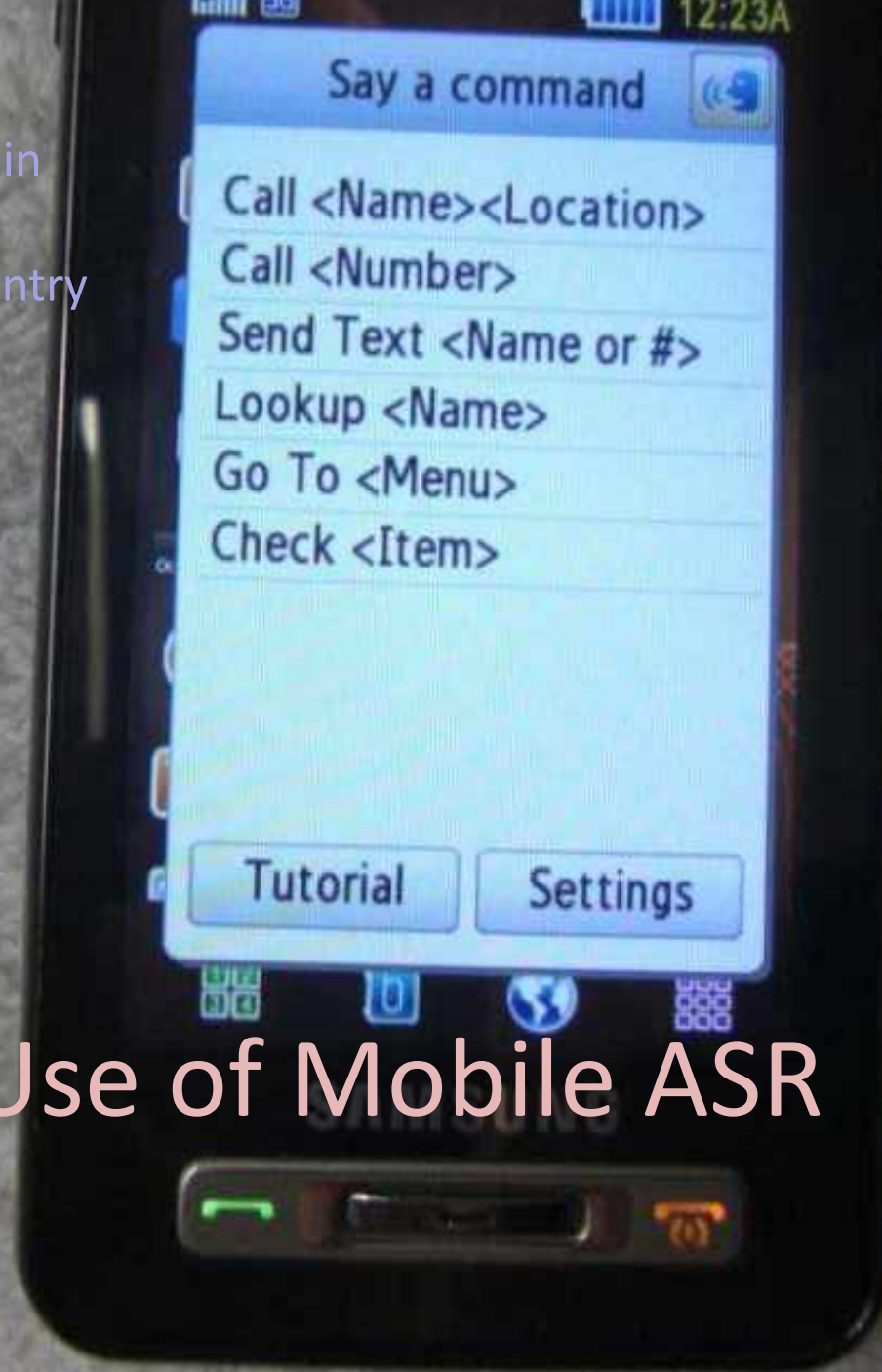
**THE STATE OF ART IS NOT BAD
AFTER ALL**

Faster than typing or navigating through different screens
As good as human most of the times

ACCEPTANCE CRITERIA

Search specific item in
a long list of objects

- ❖ phone book entry
- ❖ street names
- ❖ MP3 songs
- ❖ calendar



Use of Mobile ASR

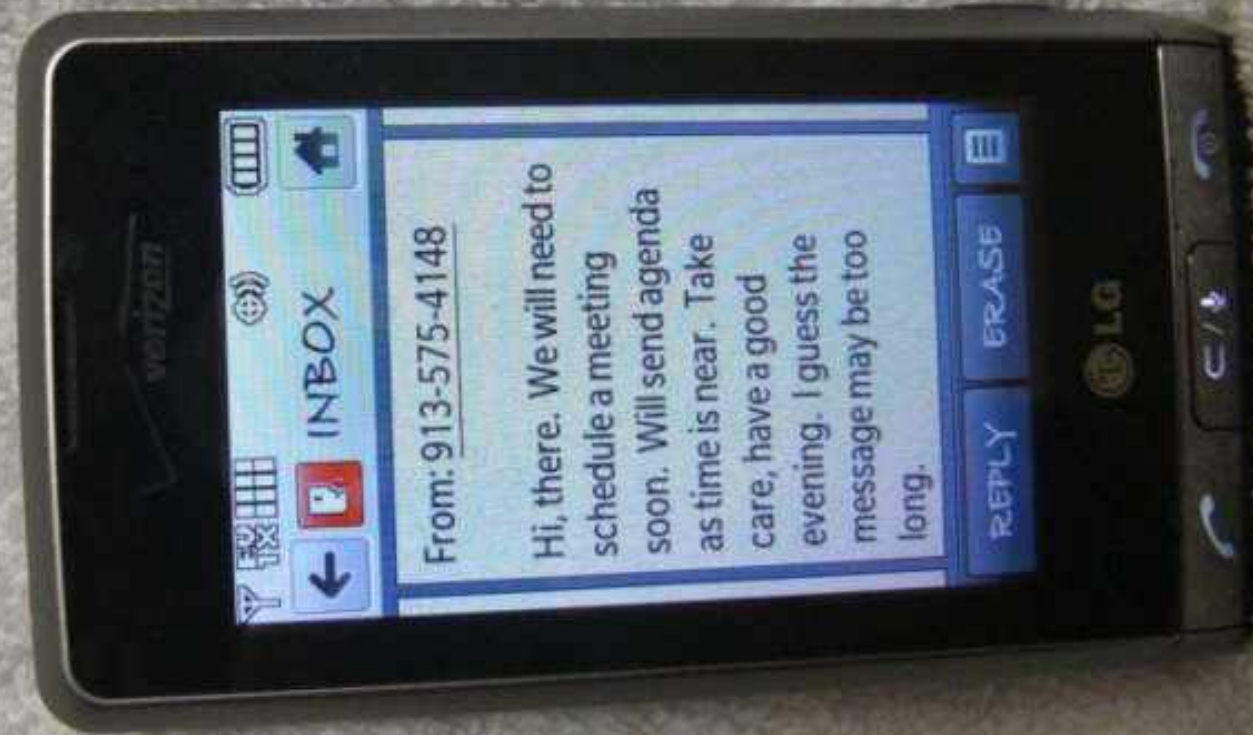


Eyesfree/Handsfree
Data Entry

Command & Control:
e.g., Voice Calls, messaging,
Menu Navigation

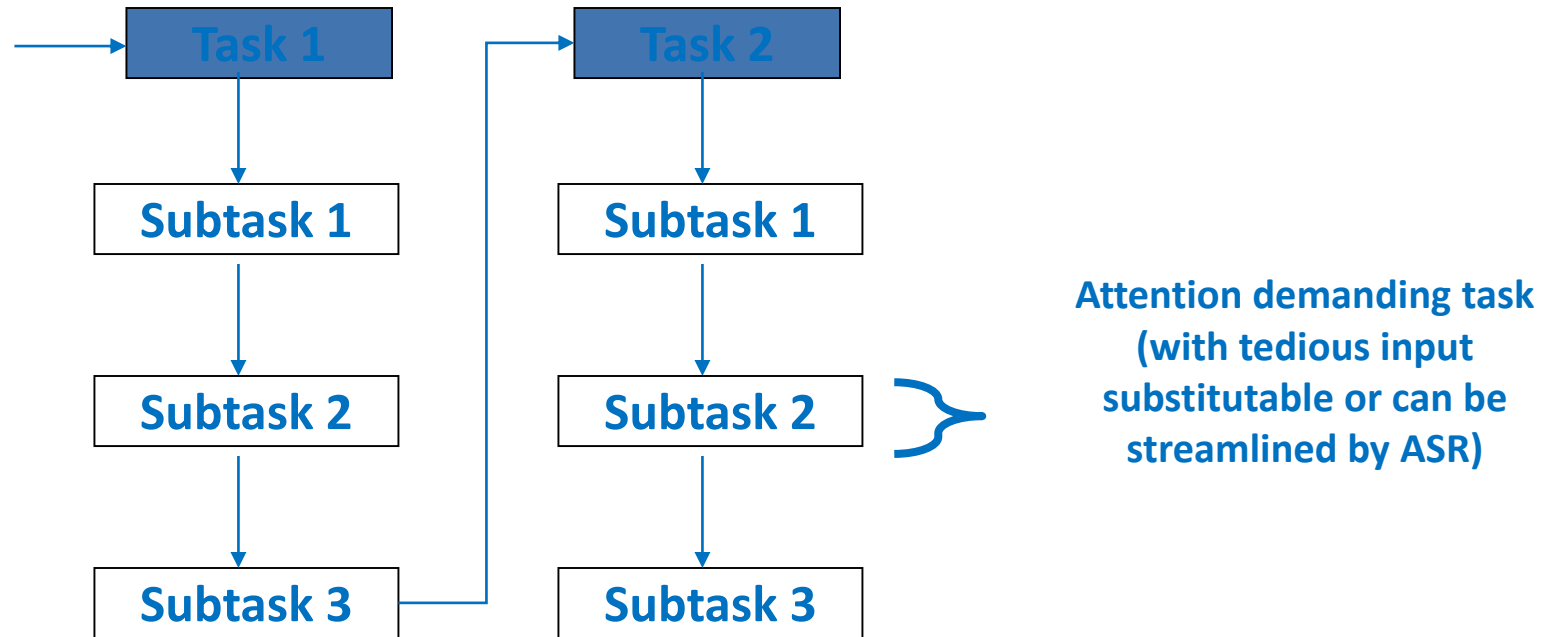
TTS Usage

Screen reading (e.g., email or text message)

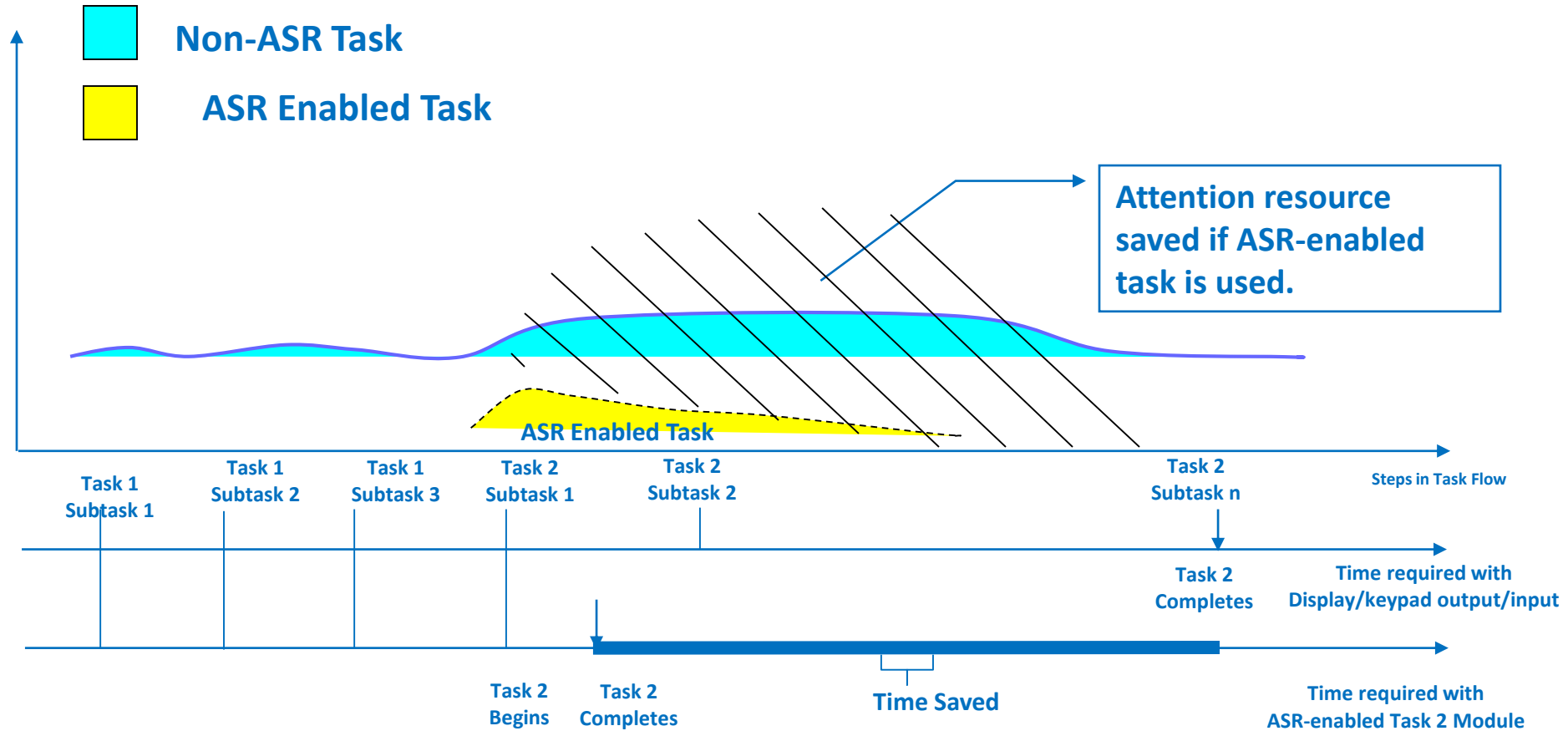


- ❖ Screen reading (e.g., email or text message)
- ❖ Menu navigation for driving or in tasks demanding high visual attention.

Attention Resource Allocation



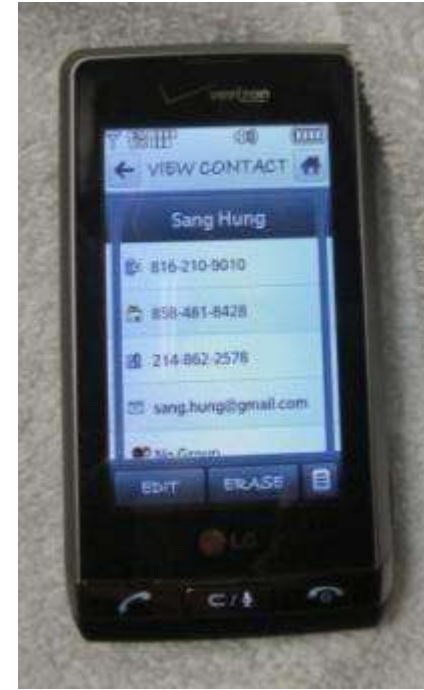
ASR-enabled task





Fine visual-motor coordination required

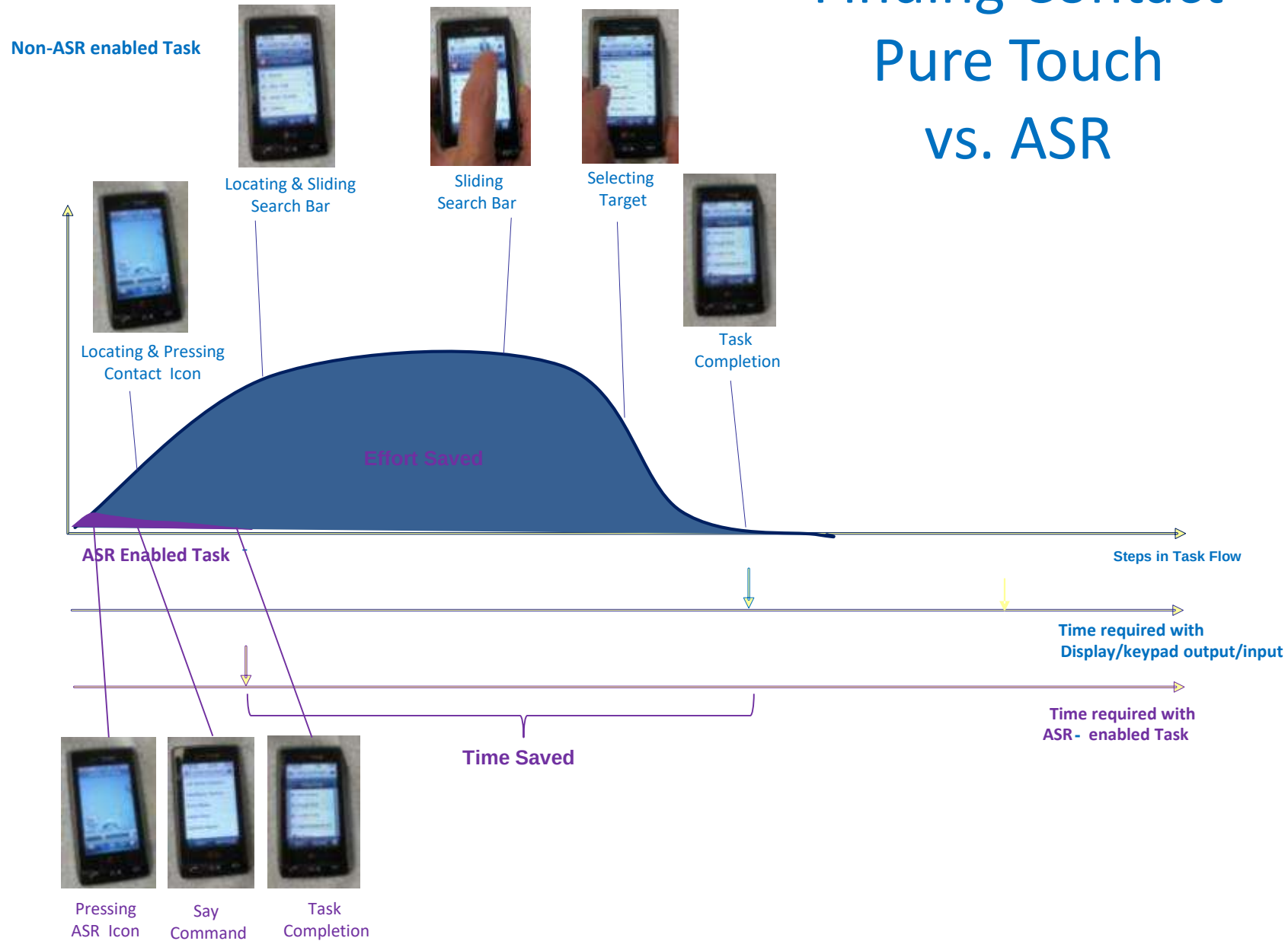
FINDING A CONTACT FROM ADDRESS BOOK WITHOUT ASR



Press physical ASR button and speak – almost no attention required.

FINDING CONTACT WITH ASR

Finding Contact Pure Touch vs. ASR

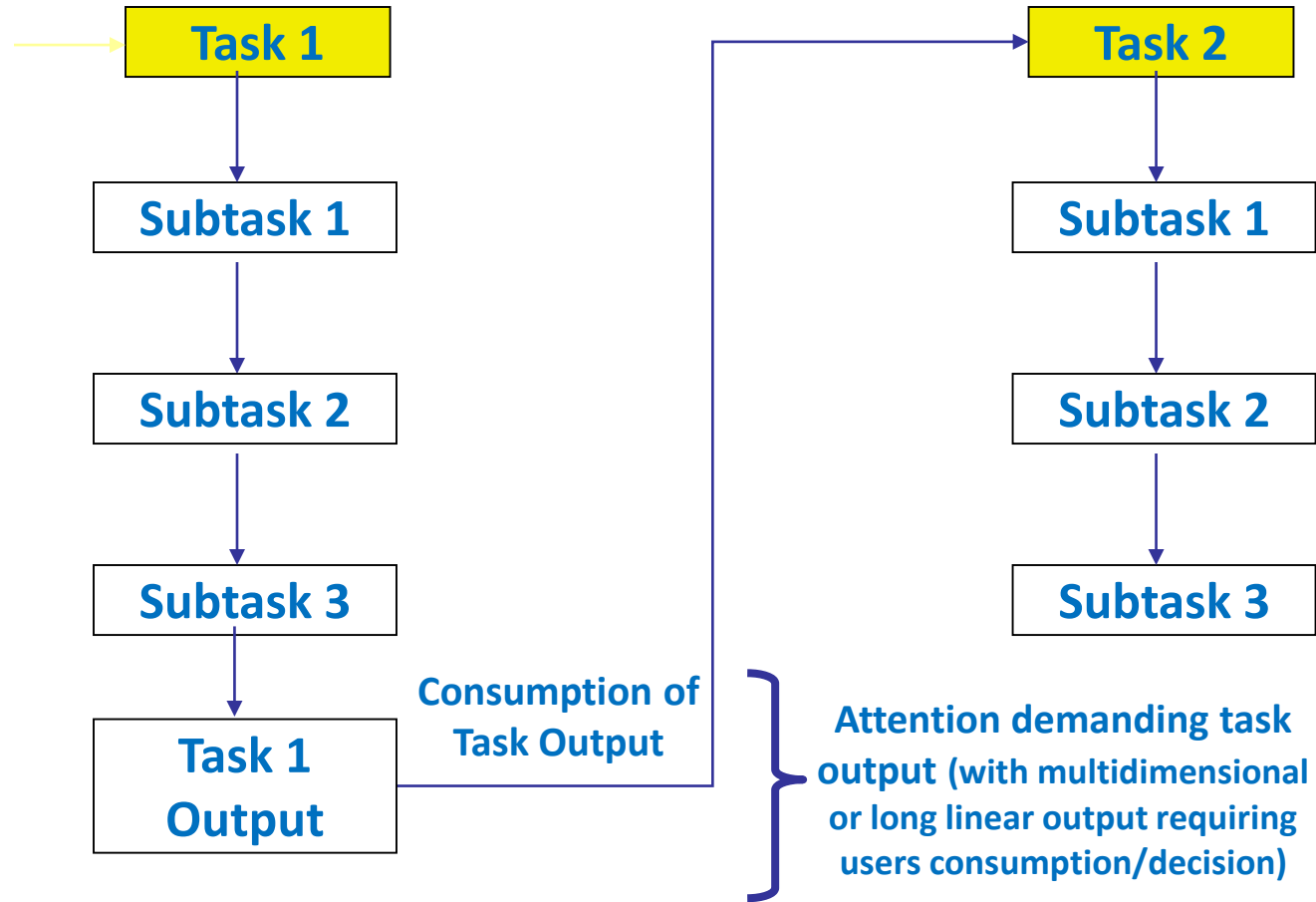


ASR/TTS UI Design Principles

- ❖ Technical (memory, processing) and human factors constraints (mobile environment) prioritize features of Mobile ASR
- ❖ Mobile ASR for voice calls
 - ❖ Optimized around making and receiving calls
 - ❖ Effortless to make calls from spoken phone number or phone book entries
 - ❖ Users should be able to say the numbers or names naturally
- ❖ Efficiency and learnability are critical to success
 - ❖ Application flow designed for efficiency and learnability
 - ❖ It should be easy to users to invoke ASR application. Direct access with a dedicated hard-key is highly recommended. This can be combined with TTS.
- ❖ User Locus of Control: Users should be able to control the application flow (e.g., for driving safety).

Attention Resource Allocation

Multimodal User Interface



ADDITIONAL PRODUCT RESEARCH SLIDES